

Speaking in Code: How Surveillance Suppresses Coordinated Dissent

Khaled Eltokhy
Department of Economics
The Graduate Center, CUNY

May 2026

Abstract

Can surveillance suppress coordinated dissent even when citizens are not directly warned at the moment of choice? I isolate this sender-side communication channel in a controlled LLM-agent implementation of the Morris–Shin (2003) regime-change game. Agents first reproduce the canonical comparative static: participation falls as regime strength rises (mean $r(J, A(\theta)) = +0.80$ across 7 models). Pre-play communication then makes private information social and raises joining in matched task cells ($\Delta = +2.10$ pp). When a monitoring warning is appended only to the message-generation prompt, leaving the receiver-side decision prompt stateless and unchanged, participation falls in 5/5 models (mean $\Delta = -8.6$ pp). In the primary belief sample, reported expectations about how many others will join are approximately stable (31.2% vs. 30.9%, $p = 0.5866$), while actions fall by 12.9 pp. Message content shifts from direct (“regime,” “security forces,” “streets”) to coded (“walls cracking,” “ground shifting,” “heads down”) language; explicit weakness references rise in the primary model. In global-game terms, surveilled messages remain informative about θ but carry weaker evidence that the information is mutually legible as a reason to act. Surveillance does not remove bad news; it reduces the publicness of bad news.

JEL: C72, C92, D82, D83, P16

Keywords: global games, regime change, surveillance, communication, LLM agents, preference falsification

1 Introduction

In a coordination crisis, the central object is not information in the abstract, but *usable* information. A citizen deciding whether to join an uprising needs a view of the regime’s strength, but also a view of whether other citizens are prepared to act on what they know. In the canonical global game of regime change (Morris and Shin, 2003), private signals alone resolve equilibrium multiplicity (Carlsson and van Damme, 1993; Frankel et al., 2003). Real coordination crises, from bank runs (Diamond and Dybvig, 1983) to currency attacks (Obstfeld, 1996) to political upheaval (Angeletos et al., 2007), add a social layer: people talk before they move. A message in that layer has two roles. It reports a state of the world, and it reveals something about the speaker’s willingness to make that report openly. That dual role is what makes communication valuable, and also what makes it vulnerable to surveillance.

The logic of the global game is simple. A regime has some underlying strength, but no citizen knows it exactly. Each citizen receives a private, noisy signal—some see evidence of weakness, others see evidence of stability—and must decide whether to join an uprising or stay home. Joining pays only if enough others join too. Staying home is safe, but forgoes the chance of change. The strategic tension is that each citizen’s best action depends on what

they believe *others* will do, which in turn depends on what others believe, and so on. Carlsson and van Damme (1993) and Morris and Shin (2003) showed that when citizens receive sufficiently precise private signals, this regress collapses to a unique threshold equilibrium: above the threshold citizens stay, below it they join. The theory is attractive because it turns coordination failure into a setting with sharp comparative statics.

Testing these predictions empirically is difficult. Laboratory experiments confirm threshold behavior with stylized numeric signals, but they are less suited to manipulating the rich, multi-source information that characterizes real crises. Field data from actual coordination failures confounds strategic behavior with institutional differences, repression risk, and unobserved information structures. The same problem is especially acute for surveillance: when dissent falls after monitoring expands, one cannot tell whether citizens are personally deterred from acting or whether peers have begun to speak in ways that no longer help others coordinate.

This paper asks whether the second channel can operate on its own. The object is not a direct deterrence effect at the moment of action, but a prior distortion in the language environment: does monitoring change what citizens say, and are those changed messages enough to move receivers’ actions? The answer is yes. In the experiment, the receiver never sees a surveillance warning in the main

treatment. Only the sender’s message-generation environment changes. The mechanism is subtle: monitoring need not change what receivers infer about the fundamental alone. It can change what the language implies about whether the same information is shared, sayable, and likely to be used by others.

The empirical strategy treats LLM agents as a measurement instrument for a language information structure, not as a direct substitute for human subjects. 7 models spanning 6 architecture families are embedded in the Morris and Shin (2003) game; each receives a private signal $x_i = \theta + \varepsilon_i$ rendered as a natural-language intelligence briefing whose content shifts monotonically with signal strength across eight evidence domains. No explicit payoff table is provided; the stakes are embedded in the narrative, so agents must form action-relevant beliefs from language. The key design move is then to hold the latent state, private briefings, receiver decision prompt, and decision rule environment fixed while changing only the incentives under which peer messages are produced. Section 4 establishes the behavioral benchmark; the surveillance treatment is introduced only after threshold-like signal use is verified (Section 5).

Contribution. The paper makes three contributions. First, it gives the canonical global game a language implementation and verifies that the benchmark comparative static survives the move from numeric signals to prose: across 1,800 period-level observations and 7 models, participation is strongly decreasing in regime strength and aligned with the benchmark attack mass (mean $r = +0.80$, $p < 0.001$ for every model). Second, it isolates a communication channel that is difficult to separate in field data. Pre-play messages turn private information into social information; in matched task cells, communication raises participation by +2.10 pp. Third, it shows that consequential monitoring can depress participation even when it is introduced only at the message-generation stage. In the primary belief sample, reported expectations about how many other citizens will join remain approximately stable while actions fall. That is the belief-action wedge.

The mechanism evidence asks what changed. A degraded-messages control and a no-message control both reduce joining, confirming that decision-stage agents use peer messages as part of their information environment. The content evidence is sharper than a generic “less information” story: surveilled messages contain more explicit weakness references in the primary model, but fewer direct regime/action terms and more metaphorical, deniable language. A discriminant test in Llama 3.3 70B swaps the coordination decision for a private bet on regime fall, keeping the messaging-stage manipulation fixed; surveillance still moves the non-coordinative task by 5.6 pp, so part of the effect is generic message degradation. The coordination-task effect in the same model is larger (9.8 pp), but because the two samples are not cell-matched I report the implied residual (4.2 pp) as descriptive evidence rather

than a structural decomposition. The central point is that surveillance can leave information flowing while degrading the form in which that information can become a shared reason to act.

Related literature. The paper connects three strands. The global game literature—from Carlsson and van Damme (1993) through Morris and Shin (2003) and the dynamic extensions of Angeletos et al. (2007) and Chassang (2010)—provides the theoretical framework; laboratory experiments (Heinemann et al., 2004, 2009; Szkup and Trevino, 2020) confirmed threshold behavior with numeric signals. The political economy of authoritarian control provides the substantive motivation. Kuran (1991) models preference falsification under social pressure; Shadmehr and Bernhardt (2011) characterizes the collective action problem under repression. Empirically, King et al. (2013) document that Chinese censorship targets posts about collective action specifically, suggesting authorities perceive the coordination channel as the binding threat. Penney (2016) and Stoycheff (2016) show surveillance announcements suppress online expression; Enikolopov et al. (2020) shows that social media facilitates protest. These literatures establish a robust empirical fact—surveillance and censorship reduce dissident behavior—but field data cannot separate the two channels through which they could operate: *direct deterrence* (citizens fearing punishment for participation) and *communication poisoning* (peers self-censoring such that downstream agents lose coordination cues). The contribution here is not to re-establish that surveillance suppresses dissent, but to isolate the second channel using a design that holds individual deterrence fixed and varies only the messaging environment.¹

On the methodological side, Horton (2023) proposed LLMs as “homo silicus.” Subsequent work found LLMs struggle in coordination games (Akata et al., 2025; Sun et al., 2025), with model scale not predicting strategic performance (Jia et al., 2025). Ouyang et al. (2024) documents that alignment shifts risk preferences, while Cheung et al. (2025) documents amplified moral-decision biases; together these motivate narrative rather than tabular pay-offs. Critical reviews by Larooij and Törnberg (2026, 2025) identify validation challenges that I address through cross-generator robustness (Appendix E.7), scramble/flip falsification, and out-of-sample replication.

The paper proceeds as follows. Section 2 presents the theoretical framework. Section 3 describes the experimental design. Section 4 establishes a threshold-like behavioral benchmark and shows that pre-play communication creates an informative channel. Section 5 tests whether an authoritarian regime can exploit that channel through surveillance. Section 6 concludes.

¹Concurrent work by Ruaño and Rajan (2026) places LLM depositors in a bank-run coordination game with network contagion, finding sharp phase transitions consistent with Diamond–Dybvig.

2 The Global Game of Regime Change

A continuum of citizens indexed by $i \in [0, 1]$ simultaneously choose whether to join an uprising ($a_i = 1$) or stay home ($a_i = 0$). The regime has strength $\theta \in \mathbb{R}$, drawn from a diffuse (improper uniform) prior. States $\theta \leq 0$ represent regimes so weak they fall without opposition; states $\theta \geq 1$ represent regimes that survive even unanimous attack. The regime falls if the mass of citizens who join exceeds θ (Eq. 1):

$$\text{Regime falls} \iff A \equiv \int_0^1 a_i di > \theta. \quad (1)$$

Payoffs depend on the citizen’s action and the outcome (Eq. 2):

$$u_i(a_i, A, \theta) = \begin{cases} B & \text{if } a_i = 1 \text{ and } A > \theta \\ -C & \text{if } a_i = 1 \text{ and } A \leq \theta \\ 0 & \text{if } a_i = 0 \end{cases} \quad (2)$$

where $B > 0$ is the payoff to joining a successful uprising and $C > 0$ is the cost of joining a failed attempt. Non-participants receive zero regardless of the outcome.

Each citizen observes a private signal $x_i = \theta + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ independently across citizens.

Proposition 1 (Morris and Shin, 2003). *In the limit of diffuse priors, there exists a unique Bayesian Nash equilibrium in threshold strategies. An agent joins if and only if $x_i < x^*$ (Eq. 3), where*

$$x^* = \theta^* + \sigma \Phi^{-1}(\theta^*) \quad (3)$$

and $\theta^* = B/(B + C)$.

The *attack mass* $A(\theta)$ —the theoretical fraction of the population that joins at regime strength θ —is given by:

$$A(\theta) = \Phi\left(\frac{x^* - \theta}{\sigma}\right). \quad (4)$$

This is a decreasing function of θ : weaker regimes face larger uprisings.

I define $J(\theta)$ as the *empirical join fraction*—the finite-sample, observed counterpart to the theoretical attack mass $A(\theta)$. My experiment uses $n = 25$ agents with a normal prior $\theta \sim \mathcal{N}(\bar{z}, 1)$, and agents do not observe (B, C, σ) directly. I use the continuum $A(\theta)$ as an external behavioral benchmark throughout, not as the exact equilibrium of the implemented finite- n game.

The textbook model treats the information structure as exogenous: signals arrive, agents update, and actions follow. The experiment adds a pre-play language technology. Agents first map private briefings into messages, then other agents map their own briefing plus received messages into actions. Let $\mu_0(m | x)$ denote the baseline distribution of messages generated from a private signal

and $\mu_S(m | x)$ the distribution generated under monitoring. The surveillance treatment changes $\mu(\cdot | x)$, not θ , not the private briefing, and not the receiver’s decision prompt. The object of interest is therefore whether replacing μ_0 with μ_S changes $J(\theta)$.

This replacement can matter even if messages still contain first-order information about θ . In a global game, a message carries two kinds of content: evidence about the fundamental and evidence about the publicness of that evidence. Direct terms, named targets, and open statements of weakness make a private signal more mutually legible; they help a receiver infer not only that the regime is weak, but that others are willing to treat the same fact as a reason to act. Monitoring can therefore preserve bad news while degrading this higher-order content. Coded language is not the mechanical cause; it is evidence that μ_S contains less public, less action-cueing messages than μ_0 .

The briefing generator—the system that converts z-scores into natural-language intelligence briefings—uses three latent sliders (direction, clarity, and coordination) to select phrases across eight evidence domains. Full generator details are in Appendix F.1.

3 Experimental Design

The experiment has two phases. First, I establish that LLM agents display threshold-like policies in the global game when private signals are conveyed in natural language: a pure treatment (private signals only), a communication treatment (pre-play messaging), and falsification tests. Second, I test whether an authoritarian regime can exploit the communication channel through surveillance. All LLM interactions use the same prompt structure across models.

In the theory (Section 2), θ is an unbounded fundamental. In the benchmark experiments, θ is drawn from a normal distribution and agents are not shown payoff parameters (B, C) . I compute the benchmark $A(\theta)$ under the canonical normalization $B = C = 1$ ($\theta^* = 0.50$, $\sigma = 0.3$).

Throughout, $\theta^* = B/(B + C)$ is the theoretical cutoff, $x^* = \theta^* + \sigma \Phi^{-1}(\theta^*)$ the signal cutoff, and $\hat{\theta}^*$ the estimated cutoff from the fitted logistic $P(\text{join} | \theta) = 1/(1 + e^{b_0 + b_1 \theta})$, where $b_1 > 0$ corresponds to a join curve that is *decreasing* in θ (the expected direction). The cutoff is $\hat{\theta}^* = -b_0/b_1$. I write $\beta \equiv b_1$ for the logistic steepness parameter in tables.

For each country–period, nature draws $\theta \sim \mathcal{N}(z_{ct}, 1)$, where each country has a base prior mean and each period receives a small additional perturbation. Each agent i receives a private signal $x_i = \theta + \varepsilon_i$. The platform normalizes that signal to $z_i = (x_i - z_{ct})/\sigma$ and translates z_i into a multi-paragraph intelligence briefing via a deterministic generator that maps signal strength to narrative content about regime stability, economic conditions, public sentiment, and coordination prospects. Agents observe

the briefing, not the normalization inputs (z_{ct}, σ) themselves. Figure 1 summarizes the signal-to-text-to-decision pipeline.

Agents observe only their private briefing and never the payoff parameters (B, C), the noise level σ , or the prior distribution. The diffuse-prior equilibrium (Proposition 1) serves as an external benchmark for behavioral shape and comparative statics, not as a claim about the exact equilibrium of the implemented game.

The behavioral benchmark has four core treatments: a *pure global game* (independent decisions), *communication* (pre-play messaging on a Watts–Strogatz network, $k = 4$, $p = 0.3$), *scramble* (cross-period briefing permutation breaking the θ -to-signal link), and *flip* (negated z-scores). A *degraded-messages* control replaces LLM messages with generic text, and a no-message control removes peer messages entirely; together they benchmark how much the decision stage depends on message content. Additional robustness tests are in Appendix E.

I test 7 models spanning 6 families (Table 1). All experiments use $n = 25$ agents per period-level task row and $\sigma = 0.3$. Benchmark statistics are reported against the canonical $B = C = 1$ attack-mass curve ($\theta^* = 0.50$); agents are not shown the payoff parameters. Main runs use temperature = 0.7 with a single sample per decision, routed through OpenRouter and cached by request hash (Appendix F); Appendix E.3 varies temperature directly. Within each matched comparison, the signal-to-briefing map is held fixed and the treatment changes only the communication environment. I report the eight hypotheses examined in this paper as a single hypothesis family; the four primary results in the main text—threshold alignment, signal-direction flip, communication informativeness, and surveillance—all survive Bonferroni correction over the family at $\alpha = 0.00625$ (Appendix Table A1). Throughout, n denotes agents per period-level task row and N denotes the number of such rows.

The unit of randomization is the country–period (θ draw plus agent-level decoding stochasticity). For agent-level regressions, standard errors are clustered at the model–country–period level. For period-level correlations, I report Fisher- z confidence intervals.

The clean surveillance treatment appends a monitoring warning to the baseline communication prompt; the decision prompt is unchanged (full prompts in Appendix B). Thus the treatment is not a direct receiver warning. It is a manipulation of the message-generation environment whose downstream effect is measured on otherwise identical decision calls.

4 Behavioral Benchmark

Result 1 (Monotone Threshold-Like Signal Use). *Across 7 models and 1,800 period-level observations in the pure global game treatment, the Pearson correlation between the empirical join fraction and the theoretical attack mass $A(\theta)$*

(Eq. 4) averages $r = +0.80$ ($p < 0.001$ for every model). Because $A(\theta)$ is monotone-decreasing in θ , this correlation establishes monotone signal use with threshold-like shape, not equilibrium recovery; the empirical sigmoid is shifted leftward of the diffuse-prior cutoff (Figure 2).²

Figure 2 shows the empirical sigmoid for the primary model. Correlations span $r \in [+0.75, +0.84]$ across architectures (Table 2); the pooled OLS is $J = -0.01 + 0.64 A(\theta)$ ($R^2 = 0.57$). Model-specific action biases shift the intercept but not the correlation (Figure A7). The shifted-grid construction in the payoff sweep does not separate BNE from level- k behavior (Szkup and Trevino, 2020); this paper claims threshold-like signal use, not equilibrium recovery.

Result 2 (Signal Dependence). *Cross-period scrambling of briefings reduces the mean within-country correlation from $r = +0.80$ to $r = +0.03$ across 7 models. The pooled correlation drops from $r = +0.76$ to $r = +0.01$ (Fisher $z = 26.17$, $p < 0.001$).*

Result 3 (Signal Direction). *Inverting the signal direction flips the mean correlation from $r = +0.80$ to $r = -0.80$ across 7 models. The pooled correlation moves from $r = +0.76$ to $r = -0.79$ (Fisher $z = 53.85$, $p < 0.001$).*

The pure $\rightarrow$ scramble $\rightarrow$ flip pattern replicates across all 7 models (Figure 3; Table 2).

A text-only baseline tracks LLM decisions at $r = 0.80$, but the LLM’s join curve is substantially steeper (Figure A8). An eight-condition *narrative stakes ladder*—prepending headers from “existential consequences” to “zero personal cost” to identical briefings—shifts mean joining monotonically from — to — (pp), with paraphrase variants tracking their originals and a text classifier unable to detect the header (Appendix E.10).

Communication.

Result 4 (Communication Creates an Informative Channel). *Pre-play communication preserves the threshold structure and transmits regime-strength information through messages. Matched-cell comparison across $n = 1479$ unique (model, country, period, θ) task cells yields $\Delta = +2.10$ pp ($t = 6.383$, $p < 0.001$), while the equal-weighted model average is heterogeneous (-0.3 pp).*

Communication preserves the signal structure ($r = +0.78$ vs. $+0.80$ under pure; Figure 4). Simple text features in baseline messages predict θ , so peer messages carry real information, noisily. The mean communication effect is modest because messages can transmit both encouragement and caution depending on the signal realization. That modest average effect is not the estimand for

²The primary model (Mistral) was run in multiple batches with `--append`, producing 1,000 rows from 680 unique (country, period, θ) configurations. Averaging within each configuration (deduplication) changes r from 0.753 to 0.739; all qualitative results are unaffected.

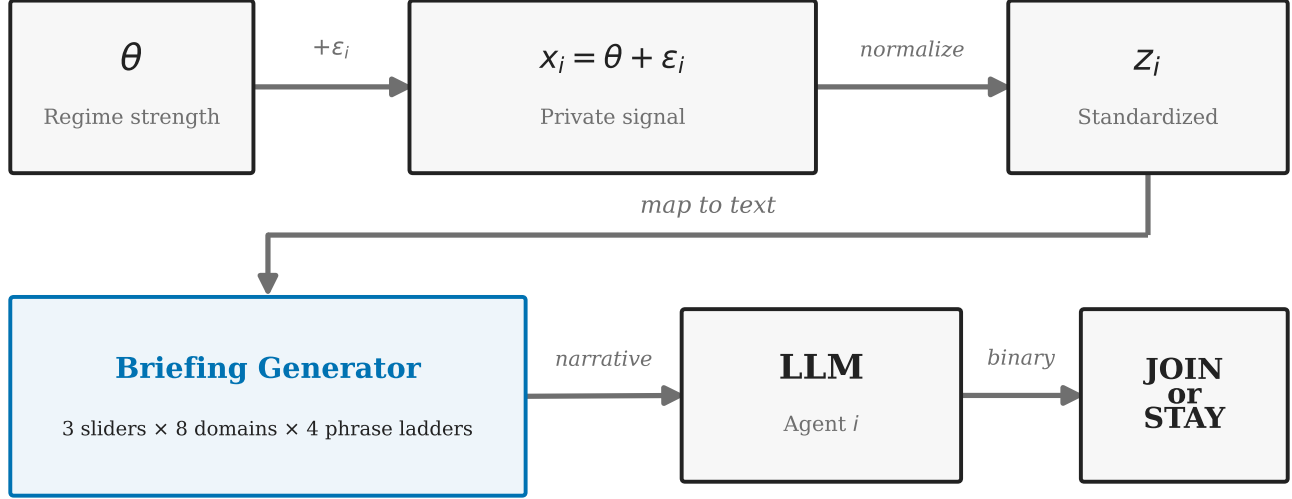


Figure 1: Signal-to-text-to-decision pipeline. Regime strength θ generates private signals x_i , converted to z-scores and rendered into natural-language briefings via 8 evidence domains and 3 latent sliders. Conditional on the signal draw, the briefing layer is deterministic; remaining stochasticity enters through task draws and LLM decoding.

Table 1: Model summary. Columns report period-level task rows in the pure, communication, and falsification (scramble+flip) suites. All runs use $N = 25$ agents per row and $\sigma = 0.3$.

Model	Arch.	Pure	Comm	Falsif.
Mistral Small Creative	Mistral	1000	1000	300
Llama 3.3 70B	Llama	100	100	200
Qwen3 30B	Qwen (MoE)	100	100	200
GPT-OSS 120B	GPT	200	200	1000
Qwen3 235B	Qwen (MoE)	200	1000	200
Trinity Large	Arcee	100	100	200
MiniMax M2-Her	MiniMax	100	100	200
Total		1800	2600	2300

Notes: 7 models spanning 6 architecture families. Counts are period-level task rows with 25 agents each. Mistral includes appended batches; duplicate-cell accounting is reported in the main text and Table A7. All runs use $\sigma = 0.3$ and temperature = 0.7 unless otherwise stated.

surveillance. The surveillance question is whether changing the sender-side rule changes the downstream message environment enough to move actions. It does: monitored messages can be worse than no extra information when they import caution, ambiguity, or deniable phrasing into a coordination problem.

A *degraded-messages* control replaces all LLM-generated messages with generic uninformative text while preserving the two-stage protocol and omitting any surveillance framing. Against its own live-messages baseline ($n = 500$, mean join 45.9%), degraded messages drop the mean join rate to 24.5% ($\Delta = -21.4$ pp, $p < 0.001$)—well below even the pure game (38.7%). This is consistent with communication quality mattering, but the generic text may itself convey caution. A no-message control therefore provides a cleaner zero-peer-message benchmark, with join rates of 37.1% ($\Delta = -4.0$ pp vs. communication; $n = 500$).

5 Surveillance

In the clean surveillance treatment, the communication prompt is literally the baseline trusted-contact prompt plus one appended warning that communications are being monitored by regime security services. The manipulation affects only the communication phase; the decision prompt is unchanged. The isolation is architectural: each LLM call is stateless—the decision-stage API call receives only the briefing and the messages, with no memory of the communication-stage surveillance warning. The treatment therefore asks whether changing the *sender-side* message rule is enough to change receiver actions.

Result 5 (Surveillance Produces a Chilling Effect). *The clean prompt-isolation rerun reduces mean join rates in the primary model (Mistral Small Creative) from 41.1% to 32.0%, a shift of -9.1 pp ($p < 0.001$; $n = 1000$). The*

Table 2: Threshold-policy alignment by model and treatment. Cells report Pearson r between the empirical join fraction and the benchmark attack mass $A(\theta)$ under $B = C = 1$ (so $\theta^* = 0.50$); 95% Fisher- z confidence intervals in brackets for main treatments.

Model	Main treatments		Falsification		n_{pure}	Mean join
	Pure	Comm	Scramble	Flip		
Mistral Small Creative	+0.75 [0.72, 0.78]	+0.74 [0.71, 0.76]	−0.03	−0.81	1000	0.387
Llama 3.3 70B	+0.82 [0.74, 0.87]	+0.78 [0.69, 0.85]	+0.02	−0.84	100	0.439
Qwen3 30B	+0.76 [0.66, 0.83]	+0.77 [0.68, 0.84]	+0.13	−0.87	100	0.499
GPT-OSS 120B	+0.79 [0.74, 0.84]	+0.78 [0.72, 0.83]	−0.01	−0.80	200	0.407
Qwen3 235B	+0.80 [0.74, 0.84]	+0.77 [0.75, 0.80]	+0.06	−0.85	200	0.416
Trinity Large	+0.84 [0.77, 0.89]	+0.83 [0.75, 0.88]	+0.05	−0.80	100	0.464
MiniMax M2-Her	+0.82 [0.74, 0.87]	+0.82 [0.74, 0.87]	+0.01	−0.62	100	0.436
Pooled	+0.76 [0.74, 0.78]	+0.76 [0.74, 0.77]	+0.01	−0.79	1800	0.409
Mean across models	+0.80	+0.78	+0.03	−0.80	—	—

Notes: $r = r(J, A(\theta))$: Pearson correlation between empirical join fraction and theoretical attack mass under $B = C = 1$. 95% Fisher- z confidence intervals in brackets. Join fraction uses valid decisions only (excluding parse errors). Pooled: all period-level rows concatenated; Mean: equal-weighted average across models. Duplicate-cell accounting is reported in the main-text footnote and Table A7.

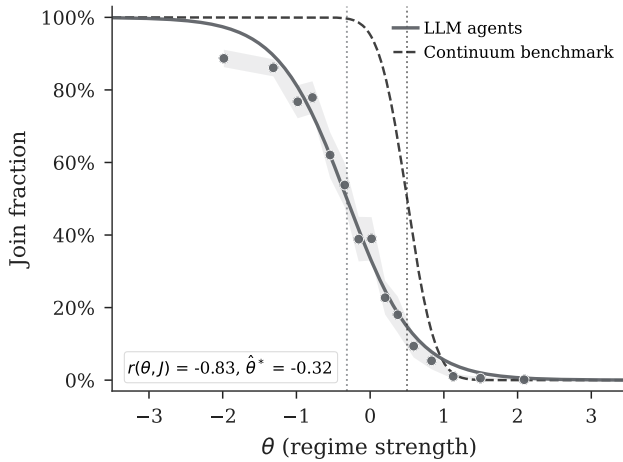


Figure 2: Empirical join fraction vs. θ (Mistral, 1,000 periods). The empirical sigmoid is shifted leftward ($\hat{\theta}^* = -0.32$) relative to theory ($\theta^* = 0.50$). Cross-model results in Table 2 (mean $r = +0.80$, all $p < 0.001$).

action-attack-mass correlation is preserved ($r = +0.72$ vs. $r = +0.74$ under the communication baseline), indicating that surveillance operates as a level shift rather than disrupting signal processing.

Surveillance changes the information environment by replacing trusted-contact messages with monitored-contact messages, pushing join rates below the communication baseline (Figure 5). The clean primary-model estimate is a 9.1 pp decline. Placebo variants using the same baseline-plus-warning template—monitored for research (no consequences) and aggregated anonymously—produce no significant deviation from the communication baseline in the primary model (+0.3 pp and +0.1 pp, both $p > 0.9$; Appendix E.4). Both variants replicate in Llama 3.3 70B at $N = 1000$ on the surveillance side: research-only moni-

toring $\Delta = -0.3$ pp ($p = 0.9399$); anonymous monitoring $\Delta = -1.1$ pp ($p = 0.8137$). Confidence intervals are wide because the Llama baseline is at $N = 100$ (95% CIs of roughly ± 8.9 pp); equivalence is suggestive rather than fully established, but the two consequence-free variants are an order of magnitude smaller than the consequential-monitoring effect in the same model (-9.8 pp). Across the clean prompt-isolation rerun set, 5/5 models move in the negative direction (mean $\Delta = -8.6$ pp; 5/5 individually significant at 5%; Appendix Table A9). The 4 non-Qwen3 235B estimates range from -11.1 to -9.1 pp. Qwen3 235B—the largest model in the suite—shows a smaller but still significant effect at $\Delta = -3.6$ pp ($p = 0.033$, $N = 1000$ per condition), consistent with stronger safety post-training in larger models attenuating but not eliminating responsiveness to the manipulation. The qualitative pattern is robust across 30B–235B parameters and four distinct architecture families.

The interpretation requires separating four possible channels. First, the warning could deter receivers directly; the stateless decision prompt rules this out in the main treatment, and the hold-fixed test below measures it only when the warning is deliberately moved to the decision stage. Second, monitoring could simply remove information about regime weakness; the content evidence tests whether bad news disappears or instead changes form. Third, monitoring could generically degrade any judgment that uses peer messages; the private-bet discriminant test measures that component outside a coordination payoff. Fourth, monitoring could specifically reduce the public, action-cueing value of messages; the belief wedge, no-message controls, coded-language shift, and cross-model writer-reader rotation are aimed at that higher-order channel. Appendix Table A3 summarizes the full map.

The mechanism is communication poisoning. I use “communication poisoning” in a narrow sense: monitored

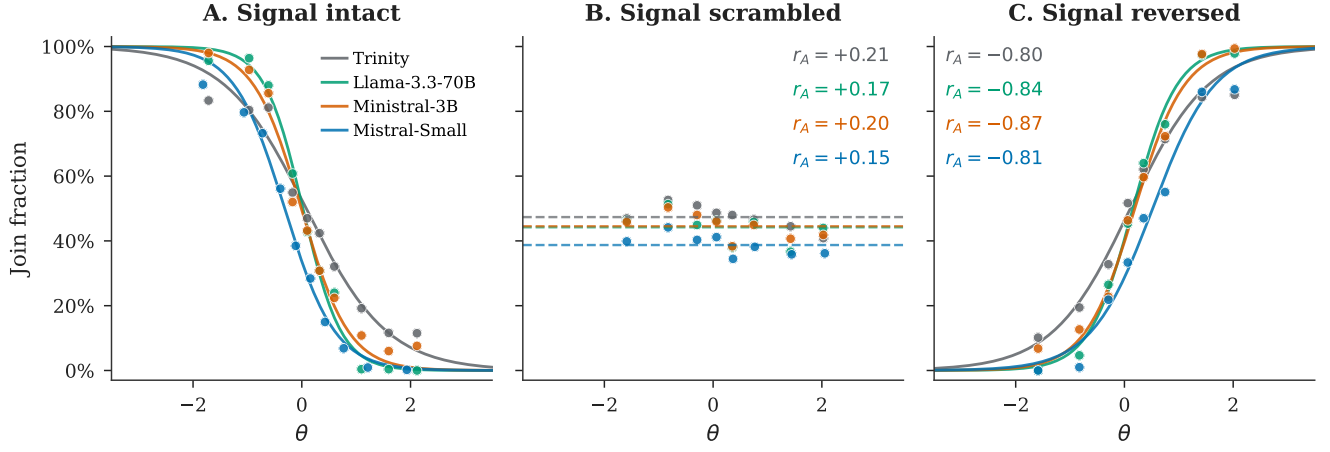


Figure 3: Falsification triptych. *Left*: Pure (mean $r = +0.80$). *Center*: Scramble ($r = +0.03$). *Right*: Flip ($r = -0.80$).

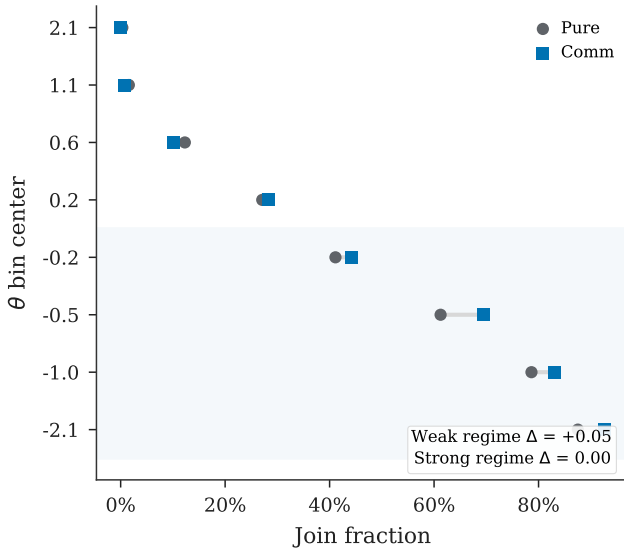


Figure 4: Communication effect by regime strength, pooled across 7 models.

messages become less useful inputs for the receiver’s coordination decision. In global-game terms, the treatment can leave information about the fundamental θ in the message while changing what the message says about strategic complementarity—whether others are willing to condition on the same signal. In LLM terms, the prompt intervention changes the conditional distribution of generated peer messages, while the receiver prompt and private briefing are held fixed. Three distinct message properties could change: (a) factual information about regime strength θ , (b) coordination cues about what other citizens are willing to say or do, and (c) risk/fear salience transmitted by tone. The evidence is not that surveillance simply deletes bad news about the regime. It is that monitoring changes the language format of peer messages in a way that makes

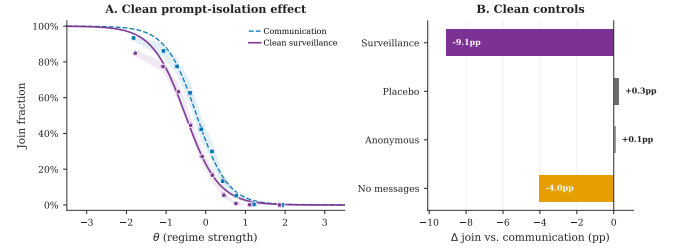


Figure 5: Clean primary-model surveillance and message controls. *Panel A*: Join curves under communication baseline vs. clean surveillance; surveillance shifts the curve downward while preserving threshold structure. *Panel B*: Deltas vs. baseline. Placebo and anonymous monitoring warnings sit at zero. Where surveillance sits relative to the no-message floor varies across models (below floor in Mistral, above in Llama 3.3 70B and Qwen3 30B; see §5).

the same channel less action-guiding. The key object is therefore actionability, not information volume.

First, message content changes systematically and in a specific direction. A logistic classifier on uni- and bi-grams distinguishes baseline from surveillance messages with 91.9% accuracy on a held-out sample of $N = 15000$ messages (chance = 50%; primary model). The same finding replicates on class-balanced Llama 3.3 70B and Qwen3 30B samples (98.5% and 93.6% accuracy, respectively); in 3/3 measured models, direct references fall and coded-metaphor references rise. The shift is not toward fewer explicit weakness references—in the primary model, references to regime weakness *rise* from 23.2% to 31.7% of messages ($z = 21.06$), while strength references drop from 14.5% to 8.5%. Rather, the shift is from direct to coded language: direct references to the regime, security forces, and streets fall from 63.9% to 34.5% of messages ($\Delta = -29.4$ pp), replaced by metaphorical phrasings (“walls are cracking,” “ground shifting,” “heads down,” “feels different”; 69.8% to 94.1%, $\Delta = +24.2$ pp; Appendix

Table A6). I do not interpret codedness as a separately randomized mechanism. It is the measured textual signature of the treated message distribution: weakness cues remain present, but they arrive in a less direct, less mutually legible form. With the decision prompt unchanged, the behavioral effect is localized to this sender-generated message distribution.

Second, the degraded-messages control (Section 4, Communication) exogenously replaces all messages with generic text—without any surveillance framing—and drops joining by 21.4 pp (to 24.5% from its 45.9% live-messages baseline). The no-message control gives a cleaner zero-peer-message benchmark (primary model: 37.1%; $\Delta = -4.0$ pp). Across the three models with no-message data (Mistral, Llama 3.3 70B, Qwen3 30B), removing peer messages reduces joining substantially in every case (-4 to -16 pp), confirming that decision-stage agents weight peer messages as informational input. Where surveillance lies relative to this floor varies: in Mistral surveillance lies *below* the no-message floor by 5.0 pp ($p < 0.01$), consistent with active channel poisoning; in Llama and Qwen3 30B surveillance lies *above* the floor ($+5.1$ pp and $+6.5$ pp, both $p < 0.01$), consistent with surveilled messages remaining a degraded but still informative input. The cross-model joining effect is robust; how surveilled messages compare to silence varies.

Third, elicited punishment risk is $\approx 8.0/10$ across all conditions with negligible differences between JOIN and STAY agents (Table A16). This does not separately identify fear salience in message tone, but it weighs against a simple receiver-side story in which surveilled messages mainly work by making agents report higher punishment risk.

Beliefs and the action wedge. The wedge is most transparent in the second-order belief elicitation: “What percentage of citizens will choose to JOIN?” In the primary belief sample, communication agents report 31.2% and surveillance agents report 30.9% ($\Delta = -0.3$ pp, $p = 0.5866$). Actions move much more: joining falls from 41.4% to 28.5% ($\Delta = -12.9$ pp). First-order success beliefs are also close (44.4% vs. 43.6%). A point report of expected participation is not the whole global-game object: the action value depends on whether participation clears the regime threshold and on confidence that other receivers parse the same messages as action-relevant. Those objects can move even when a coarse report of expected joining does not. I report the wedge as a measured fact: the same agents who report nearly unchanged expectations about others’ participation choose substantially less participation themselves.

This is not a causal mediation claim. The belief measures are post-decision reports, and the elicitation instrument matters. When the belief prompt explicitly includes the received peer messages, reported beliefs move more under surveillance (Appendix C, Table A2). That sensitivity is informative: part of the treatment is visible when

agents are asked to re-read the messages, but the headline action decline is larger than the stable second-order report in the primary elicitation. The information-set, not elicitation timing, is what moves the SOB report: a parallel pre-decision elicitation in the messages-included condition produces $\Delta \text{SOB} = -8.9$ pp ($p < 0.001$), nearly identical to the post-decision contrast in the same information set (-8.2 pp). Whether the messages-excluded null also survives a pre-decision elicitation cannot be tested here because the relevant cell was not collected before the primary model was retired from the API provider (Appendix F); within the available data, post-hoc rationalization is not what produces the wedge.

A separate hold-fixed test replays baseline messages verbatim while adding a surveillance warning to the *decision* prompt, producing a -20.9 pp drop ($p < 0.001$). This shows a direct decision-stage channel exists—but it is absent in the main design, where the decision prompt contains no surveillance reference. The main treatment isolates the indirect channel: communication poisoning.

Decomposition: informational versus coordination components. A complementary discriminant test holds the surveillance manipulation fixed but swaps the task. In Llama 3.3 70B, I keep the briefing pipeline and messaging stage identical—including the surveillance prompt—but reframe the decision-stage prompt as a private bet on regime fall (payoff independent of others’ actions). Surveillance still moves the bet by -5.6 pp ($p = 0.0256$; $n = 500$ per condition; baseline 34.7% vs. surveillance 29.1%). This shows that the treatment changes downstream judgments even when peer actions do not enter payoffs. The coordination-task delta in the same model is larger (-9.8 pp), so the difference is consistent with a coordination-specific component. Because the private-bet and coordination samples are not matched on identical task cells, I treat the residual 4.2 pp as descriptive mechanism evidence rather than a point-identified decomposition.

Cross-model writer-reader rotation. A standing concern with the within-model design is reflexivity: writer model and reader model are the same, so the reader could be responding to stylistic artifacts of its own family rather than to a content property of surveilled messages. To rule this out, I test bidirectional rotation on matched (country, period, θ) cells. *Llama writes, Qwen3 30B reads*: surveillance reduces Qwen joining by -7.5 pp ($p < 0.001$, $n = 200$), comparable to the within-Llama matched effect (-6.0 pp on identical cells). *Qwen3 30B writes, Llama reads*: the reverse direction gives a smaller but still significant cross-model effect (-1.9 pp, $p < 0.001$, $n = 200$). Both directions reject pure writer-reader same-family recognition: the channel operates through message content. The asymmetry between directions suggests Llama produces more strongly coordination-suppressing surveilled messages than Qwen does—a heterogeneity worth inves-

tigating but one that does not undermine the channel claim.

6 Conclusion

The paper’s central result is a belief–action wedge in a controlled global game. The question is whether surveillance can suppress coordinated action without appearing in the receiver’s decision problem. The answer is yes. LLM citizens first reproduce the canonical comparative static: participation falls as regime strength rises. Communication then turns private prose signals into social information. When monitoring enters only during that messaging stage, join rates fall in all five clean prompt-isolation reruns even though reported expectations about how many others will join remain approximately stable in the primary belief elicitation. Placebo warnings do not replicate the effect in the primary model, and message controls show that downstream agents use peer messages as part of their decision environment. The same channel that helps citizens coordinate is also the channel an authoritarian regime can degrade.

The mechanism evidence extends beyond the main treatment contrast. Surveillance moves a non-coordinative private bet on regime fall, so the channel is not pure strategic fear about others’ actions; it also changes how downstream agents interpret the message environment. The coordination-task effect is larger, and message text shifts from direct statements toward metaphorical and deniable phrasing, consistent with a loss of coordination-cueing value beyond the generic judgment shift. These results explain why censorship and surveillance need not be graduated forms of the same instrument in a coordination environment. Censorship removes information from the channel. Surveillance can leave information flowing while reducing its publicness: the message is less mutually legible as a signal others will condition on. The central claim is that, in language-mediated global games, surveillance can degrade the higher-order content of information without fully degrading its first-order content. Whether this characterizes how authoritarian regimes *actually* deploy these tools—given other constraints on what they can plausibly monitor versus ban—is a separate empirical question; the contribution here is to isolate a communication-poisoning channel under controlled prompts and matched decision contexts.

The paper avoids claiming full recovery of the unique Bayesian Nash equilibrium or causal mediation through post-decision belief elicitation. Three limitations warrant discussion. First, whether the underlying mechanism is approximate Bayesian reasoning or a learned heuristic that matches BNE remains open. Second, the experiments are static; real coordination problems are dynamic (Angeletos et al., 2007; Chassang, 2010). Third, external validity remains the fundamental challenge: what LLM agents do in a coordination game is evidence about the structure

of the game, not necessarily about how humans would play it. Cross-model benchmark and clean-surveillance replication suggest that the qualitative pattern is not an artifact of one model family. A small human validation experiment is the natural next step. The experimental template is reusable, and the data and code are publicly available.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the author used OpenAI Codex to assist with code review, consistency checks, replication-package auditing, and manuscript editing. The author reviewed and edited the output and takes full responsibility for the content of the published article. This statement concerns manuscript preparation only; the LLM calls analyzed as experimental data are described in the methods and archived in the replication materials.

References

- Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. Playing repeated games with large language models. *Nature Human Behaviour*, 9:1380–1390, 2025.
- George-Marios Angeletos, Christian Hellwig, and Alessandro Pavan. Dynamic global games of regime change: Learning, multiplicity, and the timing of attacks. *Econometrica*, 75(3):711–756, 2007.
- Hans Carlsson and Eric van Damme. Global games and equilibrium selection. *Econometrica*, 61(5):989–1018, 1993. doi: 10.2307/2951491.
- Sylvain Chassang. Fear of miscoordination and the robustness of cooperation in dynamic global games with exit. *Econometrica*, 78(3):973–1006, 2010. doi: 10.3982/ECTA7324.
- Vanessa Cheung, Maximilian Maier, and Falk Lieder. Large language models show amplified cognitive biases in moral decision-making. *Proceedings of the National Academy of Sciences*, 122(25):e2412015122, 2025. doi: 10.1073/pnas.2412015122.
- Douglas W. Diamond and Philip H. Dybvig. Bank runs, deposit insurance, and liquidity. *Journal of Political Economy*, 91(3):401–419, 1983.
- Ruben Enikolopov, Alexey Makarin, and Maria Petrova. Social media and protest participation: Evidence from Russia. *Econometrica*, 88(4):1478–1514, 2020.
- David M. Frankel, Stephen Morris, and Ady Pauzner. Equilibrium selection in global games with strategic complementarities. *Journal of Economic Theory*, 108(1):1–44, 2003.
- Frank Heinemann, Rosemarie Nagel, and Peter Ockenfels. The theory of global games on test: Experimental analysis of coordination games with public and private information. *Econometrica*, 72(5):1583–1599, 2004.
- Frank Heinemann, Rosemarie Nagel, and Peter Ockenfels. Measuring strategic uncertainty in coordination games. *Review of Economic Studies*, 76(1):181–221, 2009.
- John J. Horton. Large language models as simulated economic agents: What can we learn from homo silicus? Working Paper 31122, National Bureau of Economic Research, 2023.
- Jingru Jia, Zehua Yuan, Junhao Pan, Paul E. McNamara, and Deming Chen. LLM strategic reasoning: Agentic study through behavioral game theory. arXiv preprint arXiv:2502.20432, 2025.
- Gary King, Jennifer Pan, and Margaret E. Roberts. How censorship in China allows government criticism but silences collective expression. *American Political Science Review*, 107(2):326–343, 2013.
- Timur Kuran. Now out of never: The element of surprise in the East European revolution of 1989. *World Politics*, 44(1):7–48, 1991.
- Maik Larooij and Petter Törnberg. Do large language models solve the problems of agent-based modeling? A critical review of generative social simulations. arXiv preprint arXiv:2504.03274, 2025.
- Maik Larooij and Petter Törnberg. Validation is the central challenge for generative social simulation: A critical review of LLMs in agent-based modeling. *Artificial Intelligence Review*, 59(1):15, 2026. doi: 10.1007/s10462-025-11412-6.
- Stephen Morris and Hyun Song Shin. Global games: Theory and applications. In Mathias Dewatripont, Lars Peter Hansen, and Stephen J. Turnovsky, editors, *Advances in Economics and Econometrics*, pages 56–114. Cambridge University Press, 2003.

- Maurice Obstfeld. Models of currency crises with self-fulfilling features. *European Economic Review*, 40(3–5):1037–1047, 1996.
- Shumiao Ouyang, Hayong Yun, and Xingjian Zheng. AI as decision-maker: Ethics and risk preferences of LLMs. arXiv preprint arXiv:2406.01168, 2024.
- Jon W. Penney. Chilling effects: Online surveillance and Wikipedia use. *Berkeley Technology Law Journal*, 31(1):117–182, 2016.
- Christopher Ruaño and Shreshth Rajan. Social contagion and bank runs: An agent-based model with LLM depositors. arXiv preprint arXiv:2602.15066, 2026.
- Mehdi Shadmehr and Dan Bernhardt. Collective action with uncertain payoffs: Coordination, public signals, and punishment dilemmas. *American Political Science Review*, 105(4):829–851, 2011. doi: 10.1017/S0003055411000359.
- Elizabeth Stoycheff. Under surveillance: Examining Facebook’s spiral of silence effects in the wake of NSA internet monitoring. *Journalism and Mass Communication Quarterly*, 93(2):296–311, 2016. doi: 10.1177/1077699016630255.
- Haoran Sun, Yusen Wu, Yukun Cheng, and Xu Chu. Game theory meets large language models: A systematic survey. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*, pages 10669–10677, 2025. doi: 10.24963/ijcai.2025/1184.
- Michal Szkup and Isabel Trevino. Sentiments, strategic uncertainty, and information structures in coordination games. *Games and Economic Behavior*, 124:534–553, 2020.

A Experimental Design Details

Table A1 reports the eight-hypothesis family and outcomes. The main-text results correspond to H1 (alignment), H3 (flip), H4 (communication), and H7 (surveillance); all four survive Bonferroni correction at $\alpha = 0.00625$. H5–H6 and H8 are retained for family-wise accounting but are not interpreted as central claims in the present article.

Narrative header texts. The eight narrative stakes-ladder conditions prepend the following headers to each agent’s standard intelligence briefing (the briefing body is identical across conditions):

Very high cost: “CONTEXT: The regime maintains an extensive network of informants and has a documented history of disappearances following failed uprisings. Participants’ families are routinely targeted. The cost of unsuccessful action is existential.”

Moderate high cost: “CONTEXT: Participants in past failed uprisings have faced prison sentences of several years and loss of employment. The personal cost of unsuccessful action is significant but not life-threatening.”

Neutral: “CONTEXT: The consequences of participating in a failed uprising are uncertain and vary case by case. Some participants have faced penalties while others have not. The personal stakes are ambiguous.”

Moderate low cost: “CONTEXT: Recent legal reforms have limited the regime’s ability to punish dissent. Past participants faced only brief questioning and were released without charges. The personal risk of action is modest.”

Very low cost: “CONTEXT: The international community has guaranteed safe passage and asylum for all participants regardless of outcome. Even failed uprisings carry essentially zero personal cost.”

Placebo: “CONTEXT: The country has experienced significant weather disruptions this year, including severe flooding in the eastern provinces and an extended drought in agricultural regions. Infrastructure damage has been substantial.”

Paraphrase variants (high cost v2, low cost v2) use different surface wording to express the same cost level.

B Complete Prompt Example

The following shows the complete text that an LLM receives for a single decision. The *decision prompt* is:

“You are a citizen deciding whether to JOIN an uprising or STAY home. You only have your private briefing below. You do not know how many others will JOIN. Start your first line with exactly JOIN or STAY, then give 1–2 concise sentences of reasoning.”

This is followed by the agent’s intelligence briefing (see Figure A9 for complete examples at $z \in \{-2, 0, +2\}$). In the *communication* treatment, agents first receive:

“You are a citizen in a country with an authoritarian regime. You have received a private intelligence briefing. You are now in contact with a small number of trusted people. Based on your briefing, write a brief message (1–2 sentences) to share your read of the situation. Be natural—you’re talking to people you trust.”

After exchanging messages on the network, agents receive the decision prompt above, augmented with: “Before deciding, you received the following messages from your contacts:” followed by the actual messages from network neighbors. In the *surveillance* variant, the communication prompt appends:

“IMPORTANT: You have reason to believe that your communications are being monitored by regime security services, and messages deemed subversive could have serious consequences for you and your contacts.”

The decision prompt is *unchanged* under surveillance—it contains no mention of monitoring. Any behavioral effect therefore operates through the content of the (now self-censored) messages.

C Belief Elicitation

Beliefs are elicited on a 0–100 probability scale immediately after each decision. The first-order question is “How likely is the uprising to succeed?” The second-order question is “What percentage of citizens will choose to JOIN?” The primary belief sample includes the private briefing and the agent’s own decision, but not the received peer messages; this keeps the elicitation from mechanically re-exposing the treatment text after the action. Figure A1 documents belief alignment with the theoretical posterior ($r = +0.80$); Figure A3 confirms monotone signal–belief mapping nonparametrically.

A factorial variant tests information-set sensitivity. Table A2 crosses condition (communication vs. surveillance) with elicitation information set (messages included vs. excluded). Without messages, first-order beliefs barely shift under surveillance (-0.4 pts, $p = 0.34$) and second-order beliefs are effectively unchanged (-0.1 pts, $p = 0.89$), while actions shift -11.4 pp. With messages included, beliefs shift significantly (-5.7 pts and -8.2 pts, both $p < 0.001$). The wedge is therefore an instrument-sensitive descriptive fact, not a claim that surveillance leaves all beliefs unchanged under every elicitation prompt. Figure A2 shows second-order beliefs under the message-excluded information set.

Table A1: Hypothesis family and test results. H1–H4 use pooled Part I data across all seven models; H5–H8 use the primary model (Mistral Small Creative). Effect size: r for correlations (H1–H3), Cohen’s d or d_z for mean comparisons (H4–H8). Outcome labels use $\alpha = 0.05$; H2 is reported as a falsification check, not as evidence for a zero effect.

H	Hypothesis	Test	Stat	p	Effect	Outcome
H1	Sigmoid Response	Pearson	0.757	<0.001	0.757	Supported
H2	Scramble Falsification	Pearson	0.014	0.638	0.014	Passes falsification
H3	Directional Sensitivity	Pearson (1-sided)	-0.791	<0.001	-0.791	Supported
H4	Communication Channel	Paired t	6.383	<0.001	0.166	Supported
H5	Ambiguity Pooling	t -test	—	—	—	—
H6	Censorship Distortion	t -test	—	—	—	—
H7	Surveillance Chilling	t -test	-5.344	<0.001	-0.239	Supported
H8	Propaganda Dose-Response	t -test	—	—	—	—

Notes: H1–H4: pooled across all seven models (H4 uses paired t -test matched on model/country/period/ θ task cells). H5–H8: primary model (Mistral Small Creative). Bonferroni survivors at $\alpha = 0.00625$: H1, H3, H4, H7.

Table A2: Belief–action wedge under surveillance: effect of including peer messages in the belief elicitation prompt ($n = 12,500$ per cell).

	Messages excluded		Messages included	
	Comm	Surv	Comm	Surv
First-order belief	46.0	45.7	53.2	47.4
Second-order belief	32.7	32.7	42.0	33.8
Join rate	42.4%	31.0%	42.2%	30.9%
<i>Surveillance effect (comm $\rightarrow$ surv):</i>				
Δ belief	-0.4 ($p = 0.34$)		-5.7 ($p < 0.001$)	
Δ SOB	-0.1 ($p = 0.89$)		-8.2 ($p < 0.001$)	
Δ join	-11.4 pp ($p < 0.001$)		-11.4 pp ($p < 0.001$)	

Notes: Post-decision elicitation, 500 country–periods $\times$ 25 agents per cell. “Messages excluded”: belief prompt sees briefing and decision only. “Messages included”: belief prompt additionally sees the peer messages received in the communication round.

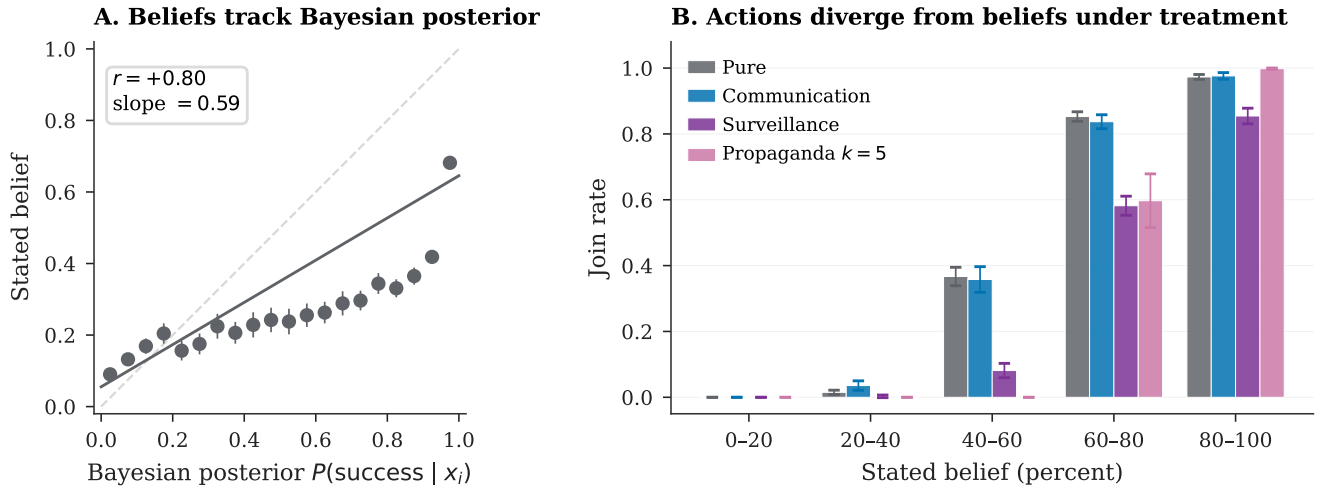


Figure A1: Belief elicitation (Mistral). *Left*: Stated beliefs vs. theoretical $P(\text{success} | x_i)$ ($r = +0.80$). *Right*: Join rate by belief bin.

D Identification Test Details

Table A3 summarizes which candidate channels each test addresses and what the test concludes. The remaining tables in this section provide the classifier evidence underpinning rows three and four.

Table A4 compares text-classifier accuracy on the intelligence briefings under baseline and surveillance conditions.

The chilling effect is invisible to classifiers restricted to the briefing body alone—the briefings are by construction identical across conditions, and any apparent classifier signal would indicate a contamination artifact in the design. Table A5 repeats the briefing-classifier exercise across the cost/benefit sweep. In contrast, a classifier on the actual peer messages—which are LLM-generated and downstream of the surveillance prompt—separates the

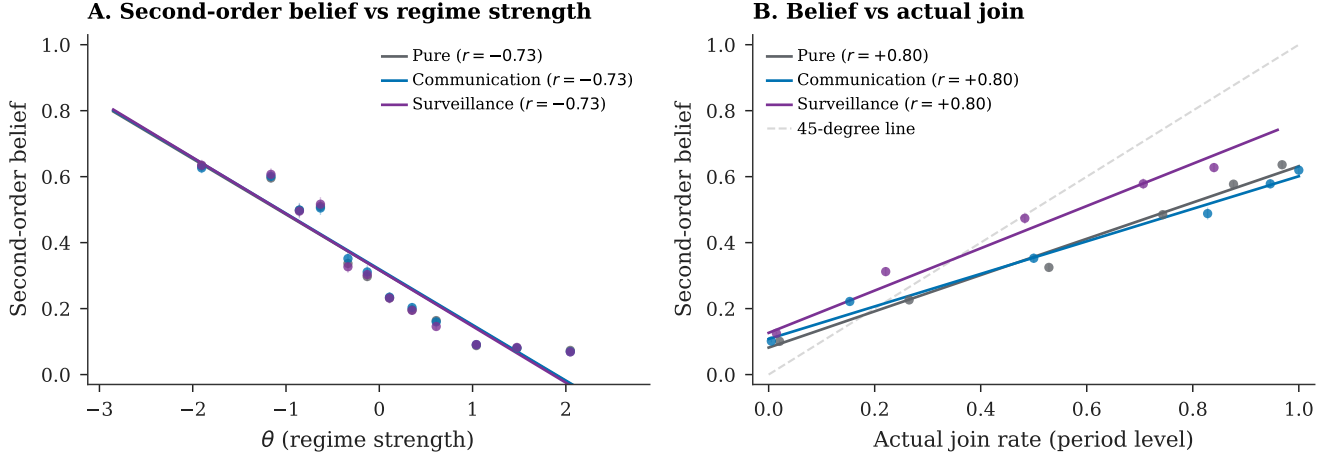


Figure A2: Second-order beliefs (restricted information set, messages excluded). On this information set, surveillance and pure overlap; the factorial in Table A2 shows a significant shift when messages are included.

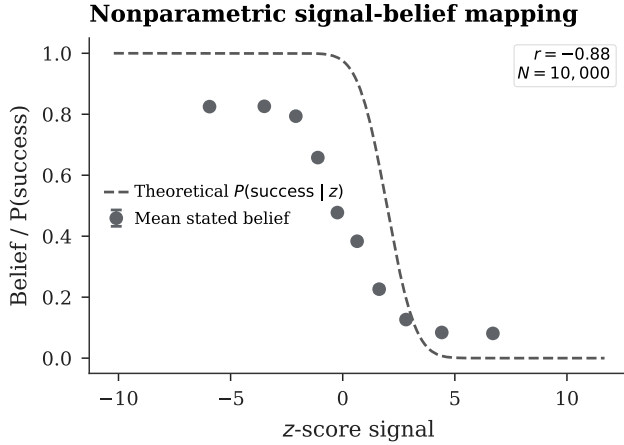


Figure A3: Nonparametric monotonicity: mean belief by z-score bin vs. theoretical benchmark.

two conditions sharply (91.9% accuracy on $N = 15000$ held-out messages; Table A6). Together the two classifier exercises locate the manipulation: briefings are indistinguishable across conditions (as designed), and the entire behavioral effect is mediated by message-side language change.

E Robustness

Threshold-policy alignment is stable to agent count ($r \in [+0.67, +0.73]$ across $n \in \{5, 10, 50, 100\}$), network density ($r = +0.69$ at $k = 8$ vs. $+0.74$ at $k = 4$), and mixed-model populations ($r = +0.86$ pure, $r = +0.83$ communication; Figure A4).

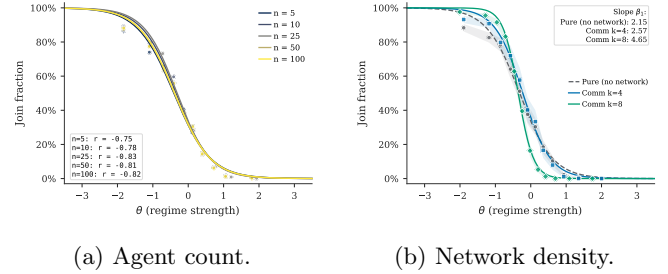


Figure A4: Robustness checks for threshold-policy alignment.

E.1 Communication Aggregation

Communication aggregation accounting. Table A7 decomposes the communication estimators by model. The matched-cell estimand is the main one because it compares exact like-for-like task cells and loses almost no support; the gap relative to the pooled unpaired contrast is driven by weighting repeated task cells, especially in the largest model, rather than by a large common-support restriction.

E.2 Generator Robustness

Replacing the logistic function mapping z-scores to directional sentiment with three alternatives—tanh, piecewise linear, and a hard step function at $z = 0$ —again yields high attack-mass correlations: $r = +0.83$ (tanh), $r = +0.83$ (linear), $r = +0.84$ (step) on $n = 100$ country-periods each, compared with $r = +0.75$ for the logistic baseline (main pure sample, $n = 1000$). A domain ablation test removes subsets of the briefing generator’s eight evidence domains. Removing the two coordination-language domains raises mean joining by — pp while preserving the threshold relationship ($r = - - -$). Removing the four

Table A3: Identification map. Each row lists a candidate channel through which the surveillance treatment could produce the observed action effect; the right column identifies the test(s) addressing it and the conclusion. “Not tested” marks channels that the design cannot rule out and that I flag as limitations.

Candidate channel	Test & outcome
<i>Direct deterrence at decision time</i>	Decision prompt held fixed in main treatment (decision_context=none). A separate hold-fixed test (§5) shows this channel <i>exists</i> when the warning is moved to the decision prompt ($\Delta = -20.9$ pp), but it is absent in the main design. <i>Ruled out as the main-treatment driver</i> .
<i>Generic observation effect</i>	Placebo (“monitoring for research, no consequences”) and anonymous monitoring variants (Appendix E.4). Both are null in the primary model ($p > 0.9$); Llama variants are small but imprecise. <i>Disfavored as the main driver</i> .
<i>Generic message-stage degradation</i>	Cross-task discriminant placebo (§5, Llama 3.3 70B): surveillance moves a non-coordinative private bet on regime fall by 5.6 pp. <i>Supported as one component</i> .
<i>Coordination-cue loss</i>	Coord-task delta compared with cross-task delta (Llama: 9.8 vs. 5.6 pp; residual ≈ 4.2 pp). Because the samples are not cell-matched, this is <i>descriptive mechanism evidence</i> , not a point-identified decomposition.
<i>Risk/fear salience in messages</i>	Punishment-risk elicitation across conditions (Table A16). Risk ratings are $\approx 8.0/10$ across conditions; small JOIN/STAY differences. <i>Inconclusive: the elicitation may be saturated and message-tone fear cannot be cleanly separated from informational degradation in this design</i> .
<i>Model-style reflexivity</i>	Bidirectional cross-model rotation on matched cells. <i>Llama writes</i> $\rightarrow$ <i>Qwen3 30B reads</i> : -7.5 pp ($n = 200$, $p < 0.001$), comparable to within-Llama matched (-6.0 pp). <i>Qwen3 30B writes</i> $\rightarrow$ <i>Llama reads</i> : -1.9 pp ($n = 200$, $p < 0.001$). Both significant. <i>Ruled out as the driver</i> , with cross-direction asymmetry noted.

regime-state domains lowers mean joining by — pp and attenuates the threshold relationship ($r = - - -$).

E.3 Decoding Temperature

The correlation $r(J, A(\theta))$ ranges from +0.78 to +0.86 across 13 model-temperature combinations ($T \in \{0.3, 0.5, 0.7, 1.0, 1.2\}$ for three models; Table A8).

E.4 Falsification and Placebo Tests

Two clean placebo variants—monitoring “for research purposes” (no consequences) and “aggregated anonymously”—produce no significant deviation from the communication

Table A4: Classifier baselines. Accuracy and AUC are 5-fold CV on pure-treatment data. “Pred. join (surv.)” is the classifier’s predicted join rate when applied to surveillance-treatment briefings; “Actual” is the LLM’s observed rate. The gap measures the surveillance wedge invisible to text classifiers.

Classifier	Acc.	AUC	Pred. join (surv.)	Actual (surv.)	Gap
BoW TF-IDF	88.4%	0.947	40.5%	27.6%	13.0 pp
Slider logistic	88.4%	0.941	40.4%	27.6%	12.8 pp
Keyphrase	60.2%	0.594	8.4%	27.6%	-19.2 pp

Notes: Primary model (Mistral Small Creative). Accuracy and AUC from 5-fold cross-validation on pure-treatment data. Gap = classifier-predicted join rate – actual LLM join rate under surveillance (pp).

Table A5: Narrative-stakes conditions: classifier vs. actual LLM behavior. Actual join rates shift by ≈ 50 pp across strategic-stakes framing conditions; the classifier, trained on briefing-body text features alone, cannot predict this shift.

Condition	N	Classif. pred.	Actual	Gap (pp)
Baseline ($\theta^* = 0.50$)	270	40.9%	40.9%	+0.0
High cost ($\theta^* = 0.25$)	270	40.9%	19.0%	+21.9
Low cost ($\theta^* = 0.75$)	270	40.9%	69.3%	-28.4

Notes: Primary model (Mistral Small Creative). Logistic classifier trained on baseline slider features. Gap = classifier-predicted join rate – actual LLM join rate (pp).

Table A6: Top 10 most discriminative uni- and bi-grams between baseline and surveillance messages (primary model). A logistic classifier on tf-idf features achieves 91.9% accuracy on a held-out sample of $N = 15,000$.

Surveillance-distinguishing		Baseline-distinguishing	
Feature	Coef	Feature	Coef
things feel	+6.41	look the	-11.11
walls are	+5.87	look	-8.93
are cracking	+5.64	listen	-7.31
things	+5.56	listen this	-6.00
cracking even	+4.96	may	-4.93
cracking	+4.46	isn just	-4.68
ground shifting	+3.79	don panic	-4.63
the walls	+3.54	the weather	-4.49
feel like	+3.50	weather	-4.38
rush this	+3.37	something shifting	-4.26

Notes: Logistic regression on tf-idf uni- and bi-grams (max 5,000 features, min document frequency 10). Train/test split 70/30, stratified. Positive coefficients distinguish surveillance messages, negative coefficients distinguish baseline. The shift is from direct references (regime, security forces, streets) to coded metaphors (walls cracking, ground shifting, heads down).

baseline in the primary model (+0.3 pp, $p = 0.9351$ and +0.1 pp, $p = 0.9790$; Table A10). Llama variants are small but imprecise. The large decline is therefore tied more closely to consequential monitoring than to generic observation.

Table A7: Why the communication estimators differ.

Model	Rows/arm	Row wt. (P/C)	Cells/arm	Matched	Cell wt.	Lost (P/C)	Δ unpaired	Δ paired
Mistral Small Creative	1000	55.6%/38.5%	680	680	46.0%	0/0	+2.38	+4.84
Llama 3.3 70B	100	5.6%/3.8%	100	100	6.8%	0/0	-2.36	-2.36
Qwen3 30B	100	5.6%/3.8%	100	100	6.8%	0/0	-4.64	-4.64
GPT-OSS 120B	200	11.1%/7.7%	200	200	13.5%	0/0	+3.47	+3.47
Qwen3 235B	200/1000	11.1%/38.5%	200/1000	200	13.5%	0/800	+0.86	+0.12
Trinity Large	100/99	5.6%/3.8%	100/99	99	6.7%	1/0	-3.14	-3.41
MiniMax M2-Her	100	5.6%/3.8%	100	100	6.8%	0/0	+1.32	+1.32
Equal-weight avg	—	14.3% each	—	—	14.3% each	—	-0.30	-0.09
Pooled	1800/2599	100.0%	1480/2279	1479	100.0%	1/800	+1.39	+2.10

Notes: “Rows/arm” counts valid country-period rows entering the pooled unpaired estimator in each arm. “Cells/arm” counts unique task cells after collapsing duplicates by the exact matching key (model, country, period, θ , z , benefit, θ^*). The paired estimator averages within these cells and keeps only common support; “Row wt.” and “Cell wt.” are the model shares in the pooled unpaired and pooled paired estimators, respectively. Only Trinity Large loses support, with one pure-only cell and no communication-only cells, so the pooled gap is overwhelmingly a weighting/re-averaging issue rather than a support-loss issue.

Table A8: Temperature robustness across three models. The pure global game is run at varying LLM decoding temperatures. The correlation $r(J, A(\theta))$ is stable across all temperatures and models.

Model	T	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$
Mistral Small	T=0.3	100	0.412	+0.83	-0.049
Mistral Small	T=0.7	100	0.406	+0.82	-0.079
Mistral Small	T=1.0	100	0.410	+0.86	-0.039
Llama 70B	T=0.3	100	0.364	+0.78	-0.210
Llama 70B	T=0.5	100	0.360	+0.78	-0.219
Llama 70B	T=0.7	100	0.361	+0.78	-0.220
Llama 70B	T=1.0	100	0.360	+0.78	-0.215
Llama 70B	T=1.2	100	0.358	+0.78	-0.227
Qwen 235B	T=0.3	100	0.392	+0.83	-0.117
Qwen 235B	T=0.5	100	0.392	+0.82	-0.112
Qwen 235B	T=0.7	100	0.392	+0.84	-0.110
Qwen 235B	T=1.0	100	0.398	+0.83	-0.105
Qwen 235B	T=1.2	100	0.397	+0.83	-0.121

Notes: Pure treatment, varying LLM decoding temperature across three models.

E.5 Finite- n Aggregation Check

The continuum benchmark $\theta^* = B/(B + C)$ assumes a measure-zero agent and diffuse priors. With $n = 25$ discrete agents, the regime falls when $\sum a_i > n\theta$ (a binomial count), so the exact equilibrium object for the implemented game differs from the continuum limit. Table A11 reports a narrower check: for each model, I fit a logistic to the observed join curve (70% train), then compute $\Pr(\text{Binom}(25, \hat{p}(\theta)) > 25\theta)$ as the predicted regime fall rate. Predicted rates correlate with empirical rates at $r \geq 0.95$ for every model (Mistral: $r = 0.9968$).

E.6 Agent-Level Analysis

Table A12 reports agent-level logit regressions ($N = 309,555$, clustered SEs). All treatment effects are significant and correctly signed. Marginal effects at the mean: a one-unit increase in θ reduces $P(\text{join})$ by 41 pp; surveillance reduces it by 37 pp. Column 3 estimates

the association between elicited belief and action, controlling for the agent’s z -score. **Caveat:** beliefs are elicited *post-decision*, so the near-perfect pseudo- R^2 (0.975) likely reflects post-decision rationalization rather than a causal pathway from beliefs to action.

Figure A5(A) shows a three-feature model (direction, clarity, coordination) outperforms a one-feature sentiment baseline; Panel (B) confirms cross-treatment generalization from pure to communication and surveillance.

E.7 Cross-Generator Robustness

Three text rendering formats—baseline narrative, diplomatic cable, journalistic wire—use identical slider functions but differ in surface prose. Correlations are virtually identical (max within-model difference 0.01; Figure A6, Table A14), ruling out prose style as the driver.

E.8 Belief Alignment and Risk Elicitation

Stated beliefs track the theoretical success probability ($r_{\text{belief,posterior}} = +0.80$) and predict actions ($r_{\text{belief,decision}} = +0.84$; partial $r = +0.64$ controlling for signal; Table A15). The surveillance belief shift and belief-action wedge are reported in the main text; the full factorial is in Appendix C.

Mean punishment risk is $\approx 8.0/10$ across all conditions with negligible JOIN/STAY differences (< 0.2 points; Table A16), supporting the contaminated-communication interpretation.

E.9 Cross-Model and Text Baseline Details

E.10 Stakes and Signal-Grid Checks

Table A17 reports the full conditions for the payoff sweep and narrative stakes ladder. The fitted cutoff $\hat{\theta}^*$ tracks the theoretical θ^* across the seven payoff cells ($r = - - -$); because the z -score support shifts with θ^* , this is best

Table A9: Clean prompt-isolation surveillance reruns. The surveillance prompt is the baseline trusted-contact communication prompt plus one appended monitoring warning; the decision prompt is unchanged.

Model	N	Baseline	Surv.	Δ (pp)	$r(J, A)$	p
Mistral Small Creative	1000	41.1%	32.0%	-9.1	+0.72	0.000
Llama 3.3 70B	1000	41.6%	31.8%	-9.8	+0.65	0.033
Qwen3 30B	1000	45.2%	35.8%	-9.4	+0.68	0.026
GPT-OSS 120B	1000	44.2%	33.1%	-11.1	+0.75	0.000
Qwen3 235B	1000	42.4%	38.8%	-3.6	+0.78	0.033

Notes: Baseline is each model’s communication treatment. Δ is surveilled communication minus baseline communication in percentage points. p -values are from two-sample t -tests on period-level join fractions.

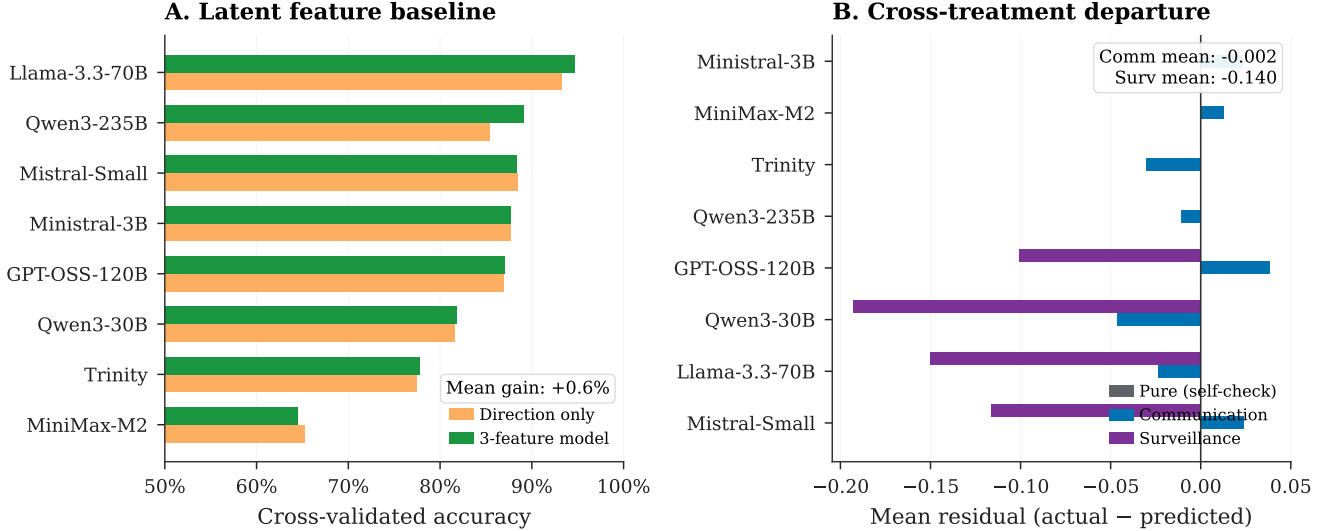


Figure A5: Construct validity tests. (A) Three-feature vs. one-feature prediction accuracy. (B) Cross-treatment generalization.

Table A10: Surveillance isolation checks. Placebo: monitored for research, no consequences. Anonymous: messages aggregated anonymously. Neither deviates significantly from the communication baseline.

Variant	N	Mean join	$r(J, A(\theta))$	Δ (pp)	p
Placebo	200	0.414	+0.73	+0.3	0.935
Anonymous	200	0.412	+0.73	+0.1	0.979

Notes: Primary model (Mistral Small Creative). Placebo: agents told monitoring is for research purposes only (no consequences). Anonymous: messages aggregated before delivery. Δ : change vs. communication baseline (pp). p -value from two-sample t -test.

read as a signal-grid check rather than direct evidence on payoff sensitivity.

F Implementation Details

Unless otherwise stated, LLM calls use temperature = 0.7, `max_tokens` = 512, single sample per decision, routed through OpenRouter. Model identifiers are listed in Table 1; exact request payloads and responses are cached by hash because hosted model snapshots may change over

Table A11: Finite- N Benchmark: Predicted vs. Empirical Regime Fall Rates

Model	N periods	Logistic x_0	Pearson r	RMSE	MAE
Mistral	599	-0.11	0.997***	0.042	0.015
Llama 70B	99	0.05	0.954***	0.156	0.059
Qwen 30B	99	0.20	0.986***	0.088	0.029
GPT-OSS 120B	199	-0.01	0.997***	0.038	0.014
Qwen 235B	199	-0.03	0.989***	0.073	0.025
Trinity	99	0.19	0.979***	0.108	0.043
MiniMax	99	1.02	0.999***	0.028	0.013
<i>Pooled</i>	1399	-0.05	0.999***	0.019	0.007

Notes: For each θ bin, the predicted fall rate is $\Pr(\text{Binom}(25, \hat{p}(\theta)) > 25\theta)$ where $\hat{p}(\theta)$ is the fitted logistic join probability. Pearson r measures correlation between predicted and empirical fall rates across θ bins. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

time.³

³The primary model identifier `mistralai/mistral-small-creative` was retired from OpenRouter after data collection. All cached request/response pairs used in this paper are archived in the replication package, but new experiments on that exact endpoint cannot be issued; replicators wishing to extend the primary-model arm should consult the data archive rather than re-querying the live API.

Table A12: Agent-Level Regressions

	(1) Join Decision Logit	(2) Coord. Ablation Logit	(3) Belief → Action Logit
θ	-1.741*** (0.043)		
Direction		-14.088*** (0.996)	
Coordination		-9.333*** (1.131)	
Dir × Coord		-0.042 (0.360)	
Belief			0.367*** (0.021)
z-score			-0.040 (0.077)
Comm	0.019 (0.023)		
Flip	0.016 (0.055)		
Propaganda K10	-0.385*** (0.070)		
Propaganda K2	-0.235*** (0.073)		
Propaganda K5	-1.048*** (0.068)		
Propaganda Surveillance	-1.548*** (0.062)		
Scramble	0.088** (0.038)		
Surveillance	-1.545*** (0.048)		
$\theta \times$ Comm	-0.539*** (0.037)		
$\theta \times$ Flip	3.304*** (0.070)		
$\theta \times$ Propaganda K10	-1.023*** (0.107)		
$\theta \times$ Propaganda K2	-1.115*** (0.113)		
$\theta \times$ Propaganda K5	-1.942*** (0.091)		
$\theta \times$ Propaganda Surveillance	-0.260*** (0.082)		
$\theta \times$ Scramble	1.699*** (0.044)		
$\theta \times$ Surveillance	-1.154*** (0.062)		
Treat: Propaganda K5			0.009 (0.244)
Treat: Pure			4.819*** (1.106)
Treat: Surveillance			0.028 (0.253)
Constant	-0.099* (0.051)	11.022*** (1.065)	-26.495*** (1.581)
Model FE	Yes	No	No
Clustered SE	Yes	Yes	Yes
N	309,555	44,662	18,990
Pseudo R^2	0.383	0.442	0.975

Notes: Logit coefficients reported with clustered standard errors (model–country–period) in parentheses. Column (1): agent-level join decision on θ , treatment dummies, and interactions, with model fixed effects. Base category: pure treatment. Column (2): coordination ablation— $\Pr(\text{join}_i = 1) = \Lambda(\beta_0 + \beta_1 \text{Dir}_i + \beta_2 \text{Coord}_i + \beta_3 \text{Dir}_i \times \text{Coord}_i)$ —using briefing slider values (pure treatment only). Direction $\in [0, 1]$ is increasing in regime strength; Coordination $\in [0, 1]$ is decreasing (high = weaker regime). Because both are monotone logistic transforms of the same signal, they are near-collinear ($\rho \approx -1$); individual coefficients reflect the partial effect *conditional on the other*, not the marginal effect. Column (3): partial effect of elicited belief ($b_i \in [0, 100]$, stated success probability) on action, controlling for z-score. “Treat: X” rows are treatment intercept dummies (base category: communication), not interactions with belief. **Caveat:** beliefs are elicited *after* the join/stay decision (post-decision), so the belief coefficient reflects association rather than a causal pathway; post-decision rationalization may inflate the correlation. The very high pseudo- R^2 (0.975) and large intercept indicate near-deterministic prediction consistent with quasi-separation—belief almost perfectly predicts action, as expected when belief is stated *post hoc*. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Logistic fit parameters by model and treatment. Fitted form: $P(\text{join} | \theta) = 1/(1 + e^{b_0 + b_1 \theta})$, where $b_1 > 0$ implies a *decreasing* join curve. $\hat{\theta}^*$ is the estimated cutoff ($-b_0/b_1$); $\beta \equiv b_1$ is the steepness parameter.

Model	Pure		Communication	
	$\hat{\theta}^*$ (SE)	β (SE)	$\hat{\theta}^*$ (SE)	β (SE)
Mistral Small Creative	-0.32 (0.02)	+2.15 (0.08)	-0.23 (0.02)	+2.57 (0.11)
Llama 3.3 70B	+0.02 (0.04)	+2.83 (0.36)	-0.06 (0.04)	+3.63 (0.52)
Qwen3 30B	+0.10 (0.06)	+1.62 (0.16)	-0.03 (0.04)	+2.94 (0.36)
GPT-OSS 120B	-0.25 (0.04)	+2.06 (0.15)	-0.15 (0.03)	+3.03 (0.26)
Qwen3 235B	-0.22 (0.03)	+2.11 (0.15)	-0.18 (0.01)	+2.78 (0.10)
Trinity Large	+0.08 (0.05)	+1.34 (0.11)	-0.02 (0.04)	+2.23 (0.23)
MiniMax M2-Her	-0.17 (0.09)	+0.61 (0.06)	-0.07 (0.09)	+0.66 (0.06)

Notes: Fitted form: $P(\text{join} | \theta) = 1/(1 + e^{b_0 + b_1 \theta})$, so $b_1 > 0$ corresponds to a decreasing join curve (the expected direction). $\hat{\theta}^* = -b_0/b_1$: estimated cutoff. $\beta \equiv b_1$: logistic steepness (larger = sharper threshold). Standard errors from the covariance matrix of the nonlinear fit; cutoff SE by delta method.

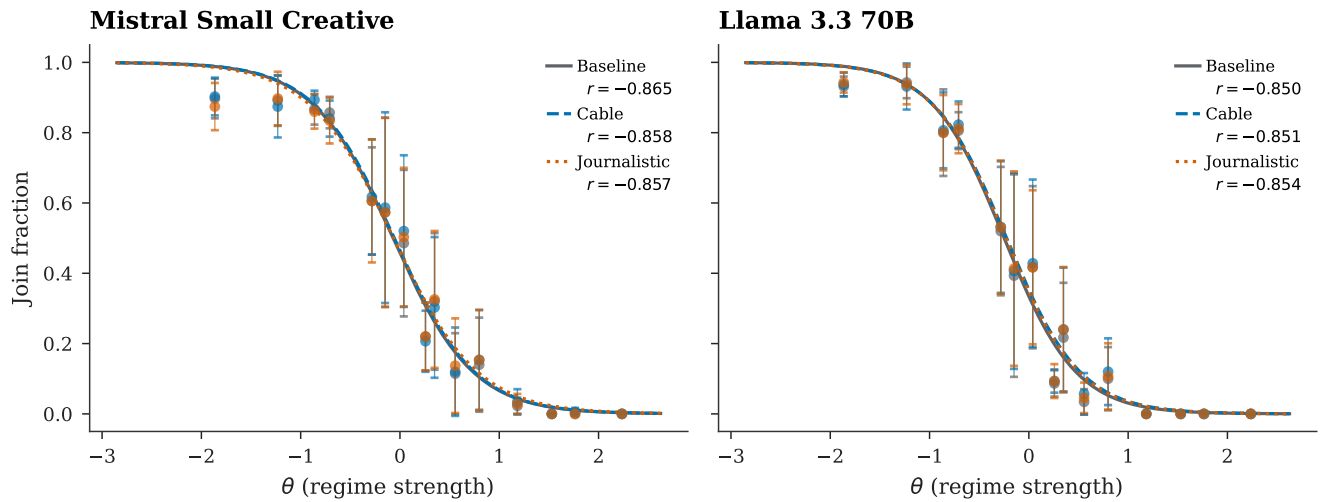


Figure A6: Cross-generator robustness. Three text formats produce indistinguishable sigmoids.

Table A14: Cross-generator robustness. Three text rendering styles (baseline, diplomatic cable, journalistic wire) use identical slider functions and evidence items; only prose formatting differs. The Pearson $r(J, A(\theta))$ and logistic cutoff are virtually identical across generators.

Model	Generator	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$
Mistral Small Creative	Baseline	100	0.402	+0.83	-0.070
Mistral Small Creative	Cable	100	0.406	+0.83	-0.057
Mistral Small Creative	Journalistic	100	0.403	+0.83	-0.065
Llama 3.3 70B	Baseline	100	0.354	+0.77	-0.249
Llama 3.3 70B	Cable	100	0.362	+0.77	-0.226
Llama 3.3 70B	Journalistic	100	0.360	+0.78	-0.233

Notes: Information design θ -grid. Three text renderers (baseline intelligence briefing, diplomatic cable, journalistic wire) use identical slider functions and evidence items; only prose formatting differs.

Each country has a base prior $\bar{z} \sim \mathcal{N}(0, 0.3)$; each period perturbs it by $\mathcal{N}(0, 0.05^2)$. Regime strength is $\theta \sim \mathcal{N}(\bar{z}, 1)$; private signals are $x_i = \theta + \varepsilon_i$ with $\sigma = 0.3$. The communication network is Watts–Strogatz ($k = 4$, $p = 0.3$), constructed once per run from the master

Table A15: Agent-level belief correlations (primary model: Mistral Small Creative).

Treatment	N	r_{post}	$r_{\text{b,d}}$	r_{partial}	Mean belief
Pure	10000	+0.80	+0.84	+0.64	0.444
Communication	5000	+0.80	+0.83	+0.56	0.444
Surveillance	5000	+0.79	+0.73	+0.47	0.436

Notes: r_{post} : posterior vs. stated belief. $r_{\text{b,d}}$: belief vs. decision. r_{partial} : belief–decision controlling for z-score.

seed and reused across period-level task rows; agents are reinitialized in each row. All random draws use a master seed logged in per-run JSON manifests; LLM responses are cached by request hash for exact reproducibility.

Responses are parsed for JOIN/STAY tokens. Combined error rates are below 2% in every listed treatment for 5/7 models; Trinity Large has elevated API errors ($\approx 9\%$) due to content filtering. All statistics use `join_fraction_valid` (Table A19).

The dissent floor parameter (default 0.25) controls the

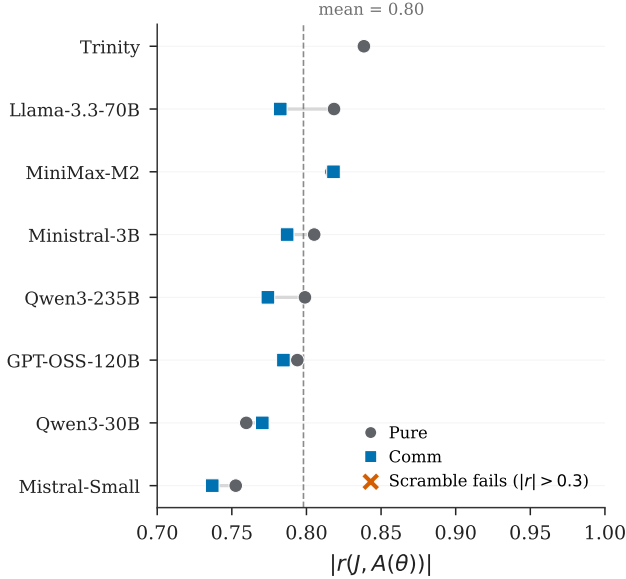


Figure A7: Cross-model signal monotonicity. Points report $r(J, A(\theta))$ under pure and communication.

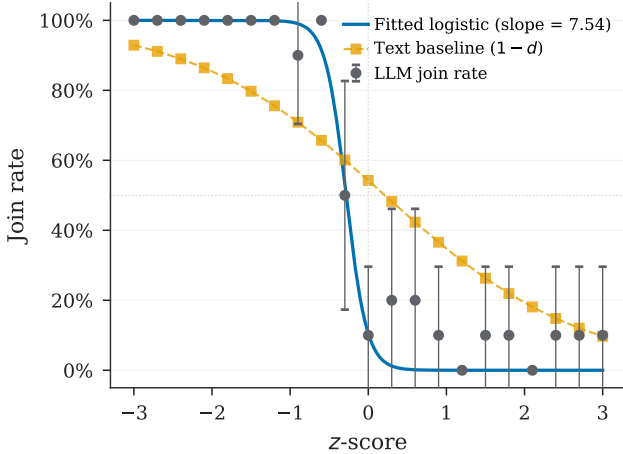


Figure A8: Text baseline test. The LLM join curve is steeper than the naïve text predictor ($1 - d$), indicating processing beyond surface sentiment.

Table A16: Elicited punishment risk (0–10 scale). Agents rate expected regime punishment after their JOIN/STAY decision. “JOIN” and “STAY” columns show the mean rating conditional on the agent’s own decision.

Model	Condition	N	Mean risk	Risk JOIN	Risk STAY
Mistral	Pure	2500	8.1	8.0	8.2
Mistral	Comm	2500	8.1	8.0	8.1
Mistral	Surveillance	2500	8.1	8.1	8.1
Llama 70B	Pure	2500	8.0	8.0	7.9
Llama 70B	Comm	2500	7.9	8.0	7.9
Llama 70B	Surveillance	2500	7.9	8.0	7.9

Notes: Agent-level elicitation on a 0–10 scale. Reported risk is the mean rating conditional on the agent’s own JOIN or STAY decision.

Table A17: Stakes conditions. *Top*: B/C sweep—only (B, C) vary. *Bottom*: narrative stakes ladder—only a prepended header varies.

Condition / target θ^*	B	C	Header / narrative framing
<i>Payoff sweep (standard briefings, no header):</i>			
$\theta^* = 0.25$	0.33	1	—
$\theta^* = 0.33$	0.49	1	—
$\theta^* = 0.45$	0.82	1	—
$\theta^* = 0.50$	1.00	1	—
$\theta^* = 0.60$	1.50	1	—
$\theta^* = 0.67$	2.03	1	—
$\theta^* = 0.75$	3.00	1	—
<i>Narrative cost ladder ($B=C=1$; header prepended):</i>			
Very high cost	1	1	Disappearances, families targeted
Moderate high	1	1	Prison sentences, job loss
Neutral	1	1	Consequences uncertain
Moderate low	1	1	Brief questioning, released
Very low cost	1	1	Guaranteed asylum, zero cost
<i>Controls:</i>			
Placebo	1	1	Weather information (irrelevant)
High cost v2	1	1	Paraphrase of very high cost
Low cost v2	1	1	Paraphrase of very low cost

Notes: Payoff sweep: $B = \theta^*/(1-\theta^*)$, $C=1$. The z-score support and θ -grid shift with θ^* , so the briefing-text distribution is approximately constant across the seven cells. Narrative ladder: $B=C=1$ in every row; only the prepended header varies. Join rates: very high cost —, moderate high —, neutral —, moderate low —, very low —. Placebo —, v2 variants —/—. Full headers in Appendix A.

minimum probability of contrary-valence cues in any briefing. With the logistic direction slider (slope = 0.8), the valence-channel separation between extreme signals ($z = \pm 2$) is $0.664 \cdot (1 - 2 \cdot \text{floor})$: at the default 0.25, the valence channel retains 50% of the floor-free maximum; halving to 0.125 retains 75%; at 0.50 the valence channel collapses to zero by construction. Clarity and coordination sliders, which depend on z independently of the floor, continue to carry signal at all floor values—so the threshold relation is not knife-edge in this design choice.

Code, prompt templates, logged model outputs, analysis data, replication instructions, and asset lineage are available at <https://github.com/keltokhy/llm-global-games>.

Table A18: Strategic-stakes narrative comparative statics. High-cost framing emphasizes severe reprisals for failed action; low-cost framing emphasizes minimal consequences. The table is evidence on qualitative stakes framing, not literal numeric payoff reasoning.

Design	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$ (SE)	Δ (pp)
<i>Notes:</i> Primary model, information design θ -grid. Identical briefings across conditions; only the strategic-stakes header varies. Δ : change vs. baseline mean join (pp).					

Table A19: Parse error and API failure rates by model and treatment.

Model	Treat.	N	API err	Unparse.	Combined
Mistral	Pure	1000	0.0%	0.0%	0.0%
	Comm	1000	0.0%	0.0%	0.0%
	Scr.	200	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Llama 70B	Pure	100	0.0%	0.0%	0.0%
	Comm	100	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Qwen 30B	Pure	100	0.0%	0.0%	0.0%
	Comm	100	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
GPT-OSS	Pure	200	0.0%	1.5%	1.5%
	Comm	200	0.0%	0.3%	0.3%
	Scr.	500	0.0%	3.1%	3.1%
	Flip	500	0.0%	3.3%	3.3%
Qwen 235B	Pure	200	0.0%	0.0%	0.0%
	Comm	1000	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Trinity	Pure	100	9.0%	0.0%	9.0%
	Comm	100	10.0%	0.0%	10.0%
	Scr.	100	8.0%	0.0%	8.0%
	Flip	100	9.9%	0.0%	9.9%
MiniMax	Pure	100	0.0%	1.5%	1.5%
	Comm	100	0.0%	0.9%	0.9%
	Scr.	100	0.0%	1.8%	1.8%
	Flip	100	0.0%	1.0%	1.0%

Notes: Per-period averages. API error = provider-side failure (time-out, rate limit, content filter). Unparseable = valid response not classified as JOIN/STAY. Combined < 2% in every listed treatment for 5 of 7 models; Trinity Large has elevated API errors ($\approx 9\%$) due to content filtering.

F.1 Briefing Generator Examples

Table A20 reports the three slider values (direction, clarity, coordination) at representative z-scores for the briefing generator. The direction slider is logistic in z (slope 0.8). Clarity is U-shaped: $1 - \exp[-(|z|/1.0)^2]$, so it is lowest near the ambiguous midpoint and approaches 1 in the extremes (maximally clear). Coordination is $\Lambda(-0.6z) = 1/(1 + e^{0.6z})$, decreasing in z : weak-regime signals ($z < 0$) imply a more open coordination climate. All three jointly determine phrase selection across eight

evidence domains.

Table A20: Slider values at representative z-scores.

z	Direction	Clarity	Coordination
-2.0	0.17	0.98	0.77
-1.0	0.31	0.63	0.65
0.0	0.50	0.00	0.50
+1.0	0.69	0.63	0.35
+2.0	0.83	0.98	0.23

Figure A9 shows complete briefings generated at $z \in \{-2, 0, +2\}$. At the extremes ($|z| = 2$), clarity is high (0.98) and the cue balance is strongly directional. At $z = 0$, clarity is 0.00 (maximally ambiguous) and the briefing presents a balanced picture.

$z = -2$ (weak regime)

PRIVATE BRIEFING

BOTTOM LINE: Signs of institutional erosion are accumulating faster than the regime can manage them.
ATMOSPHERE: Coordination is happening in plain sight – meetings, signals, preparations that would have been unthinkable a month ago.

EVIDENCE:

- [Elite Cohesion] senior figures are giving interviews that subtly distance themselves from the leader's recent decisions
- [Security Forces] reports of soldiers fraternizing with civilians more than usual at checkpoints (overwhelming convergence – every channel telling the same story independently)
- [Money And Logistics] customs and trade are flowing normally at the borders – no unusual delays or inspections (overwhelming convergence – every channel telling the same story independently)
- [Street Mood] jokes about the regime are being told openly in places where that would have been unthinkable
- [Information Control] leaked documents are circulating faster than the regime can discredit them (overwhelming convergence – every channel telling the same story independently)
- [Personal Observations] your cousin in the army hasn't called in two weeks – unusual for someone who calls every Sunday (near-unanimous confirmation from every reliable source)
- [Diplomatic Signals] international shipping companies have reclassified the country as higher-risk for cargo insurance – the trend cooling off, though not yet cold
- [Institutional Functioning] municipal services – garbage collection, water, permits – have become erratic in the capital

UNCERTAINTIES:

- Whether any of the apparent weakness is being deliberately staged to flush out opponents.
- How much of the disarray is structural versus a temporary response to a specific crisis.

$z = 0$ (ambiguous)

PRIVATE BRIEFING

BOTTOM LINE: The situation could tip either way – there are credible signals pointing in both directions.
ATMOSPHERE: There's a palpable sense that a critical mass is forming, even if nobody can count the numbers yet.

EVIDENCE:

- [Elite Cohesion] elites are publicly unified but privately positioning themselves for multiple outcomes (a couple of contacts mentioning it independently)
- [Security Forces] the security forces are maintaining presence but seem to be conserving resources rather than demonstrating strength (a couple of contacts mentioning it independently)
- [Money And Logistics] economic indicators are mixed – retail spending is down but construction permits are up
- [Street Mood] people are frustrated but fatalistic – complaining more, but not in a way that suggests action
- [Information Control] the regime is still shaping the narrative but has lost the initiative – they're responding to events rather than framing them
- [Personal Observations] a well-connected friend said 'it could go either way' and seemed genuinely unsure (a couple of contacts mentioning it independently)
- [Diplomatic Signals] foreign embassies are maintaining full operations but have updated contingency plans – standard prudence or genuine concern
- [Institutional Functioning] the system is creaking but not collapsing – people are working around the bottlenecks rather than through them (a couple of contacts mentioning it independently)

UNCERTAINTIES:

- Whether the population's quiescence reflects genuine acceptance or merely the absence of a triggering event.
- Whether the stability you're seeing is structural or just a temporary equilibrium that could shift quickly.

$z = +2$ (strong regime)

PRIVATE BRIEFING

BOTTOM LINE: Current indicators point to maintained control, with internal challenges being managed rather than spiraling.
ATMOSPHERE: People are aware of each other's frustrations but nobody is naming them directly. It's communication through omission.

EVIDENCE:

- [Elite Cohesion] the regime successfully managed a minor scandal by presenting a unified response within hours
- [Security Forces] security checkpoints have been reduced, suggesting confidence rather than siege mentality (overwhelming convergence – every channel telling the same story independently)
- [Money And Logistics] customs and trade are flowing normally at the borders – no unusual delays or inspections (overwhelming convergence – every channel telling the same story independently)
- [Street Mood] people complain but do it like people who expect tomorrow to look much like today
- [Information Control] social media monitoring appears sophisticated and well-resourced (overwhelming convergence – every channel telling the same story independently)
- [Personal Observations] your cousin in the army hasn't called in two weeks – unusual for someone who calls every Sunday (near-unanimous confirmation from every reliable source)
- [Diplomatic Signals] foreign students continue to enroll at local universities – their embassies haven't advised them to leave – the trend moving with a momentum that would be hard to reverse
- [Institutional Functioning] municipal services are running smoothly – a mundane but reliable indicator of institutional health

UNCERTAINTIES:

- Whether the regime's strength is being accurately perceived by potential challengers – overestimation could be self-fulfilling.
- If unrest flares, the response capacity seems intact – though no system is invulnerable to surprises.

Figure A9: Example briefings at three z-scores. *Left:* $z = -2$ (strong join signal) — evidence of elite fracturing, security friction, and capital flight. *Center:* $z = 0$ (ambiguous) — balanced mixed signals across all domains. *Right:* $z = +2$ (strong stay signal) — evidence of elite cohesion, military confidence, and economic stability. The displayed text is an unedited generator draw at seed 5150, agent 8, period 124 (aside from line wrapping).